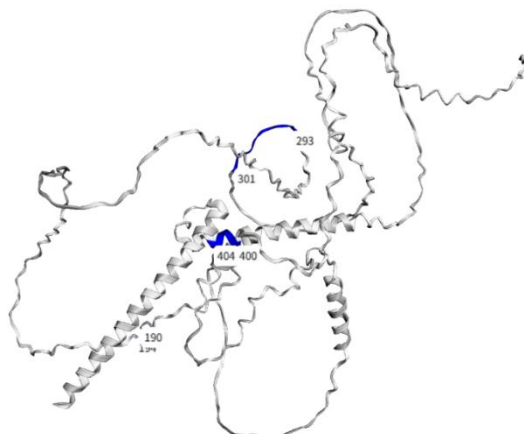
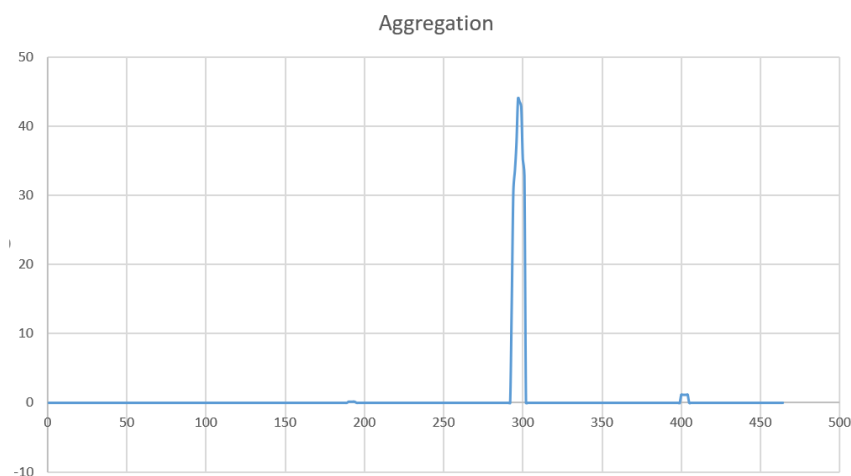


Result

We utilized TANGO software to analyze the aggregation-prone regions (APRs) of the N-MYC protein, integrating these findings with AlphaFold-generated structural models as illustrated below (Blue regions indicate the APRs).



Structure of N-MYC predicted by AlphaFold



TANGO score of N-MYC per residue

Based on the TANGO scoring results, we identified two Aggregation-Prone Regions (APRs) in N-MYC: residues 293-301 and residues 400-404. However, the 400-404 region was excluded from consideration due to its excessively low score (according to TANGO criteria, a continuous stretch of five or more residues with a score above 5 is a strong predictor of aggregation) and its predicted α -helix structure in AlphaFold.

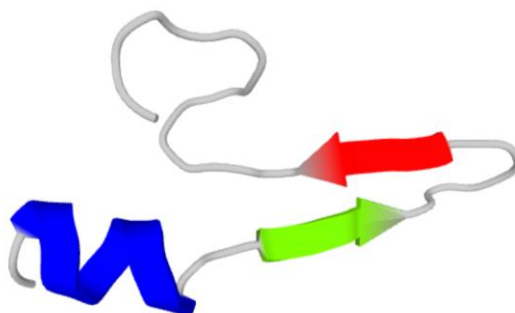
Based on the identified APRs of N-MYC, several functional peptides (FuncPepts) were designed and further evaluated using the WALTZ scoring algorithm. This screening process allowed us to isolate the candidate peptide with the highest score.

Sequence	Average Score
TFITITKGTTFTITK	93.31104
TFITITGKTFTITK	93.31104
KAVTFTIGKTFTITK	91.44206
TFITITKGAVTFTIK	90.2592
TFITITGKAVTFTIK	90.15468
KAVTFTIKGTTFTITK	90.06492
TFITITKGAVTFTK	89.47603
TFITITGKAVTFTK	89.36455
KAVTFTIGKAVTFTIK	88.81457
TFITITGKTFTITVK	88.04348
TFITITKGTTFTITVK	88.04348
KAVTFTIGKAVTFTK	88.03856
TFITITKGTTFTIVK	87.80379
KAVTFTIKGAVTFTIK	87.79264
TFITITGKTFTITVK	87.60312
KAVTFTIKGAVTFTK	86.95652
KAVTFTIGKTFTITVK	86.86362
KAVTFTIGKTFTITVK	86.48436
KAVTFTIKGTTFTITVK	85.56299
KAVTFTIKGTTFTITVK	85.48102
KAVTFTKGTTFTITVK	84.44816
KAVTFTGKTFTITVK	84.26003
TFITITVKGTTFTITVK	82.88043
KTFITITVKGTTFTITVK	82.79682
TFITITVGKTFTITVK	82.69231
KTFITITVGKTFTITVK	82.608

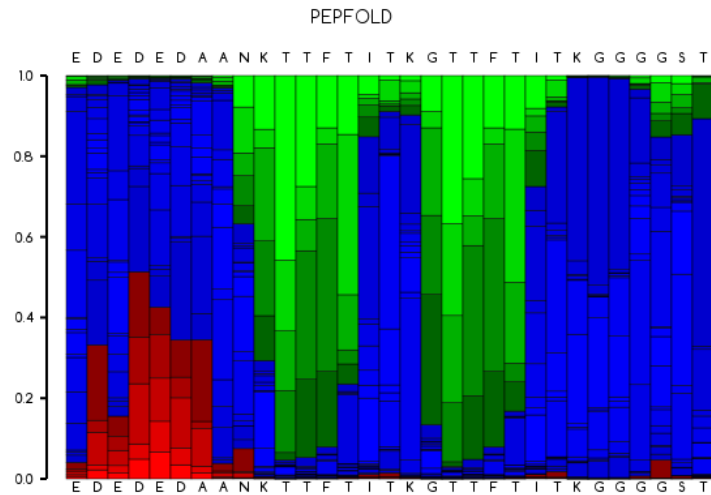
WALZE scores of peptides

According to the WALTZ scoring results, the highest-scoring peptide segment achieved a score of ~93.3, indicating that our designed short peptide possesses a strong ability to induce aggregation.

Following the structural refinement of the lead peptide, we performed structural prediction using PEP-FOLD (results shown below).



Structure of the peptide



Predicted Local Structure

In the PEP-FOLD structure predictions, the negative-charge shielding sequence tends to form an α -helix, while our designed FuncPept sequence tends to form a β -sheet, with the linker region presenting as a flexible chain. This perfectly aligns with our original design intentions and demonstrates the functional efficacy of our designed peptide in in silico simulations.

Experimental validation of the protein expression within cell models is currently pending.